Causal Dialog Route Reranker

SE team   Alternative DialNav Challenge entry, August 2026

Method and communication boundary. We submit a single-rollout dialog-navigation method built on the official RAINbow system. Only the Navigator owns and invokes the frozen LANA question generator (QG). At a dialog step, the Navigator describes its current observation as a natural-language question. The Guide, which has no QG module, localizes that question with frozen GTL and generates a natural-language route answer with the frozen LANA answer generator. Only the question and answer strings enter the Navigator's instruction state.

Our addition is Guide-private temporal localization. If the previous Guide answer was generated from route R at step t_0, the expected location at step t is $e_t = R_{\min(\max(t-t_0,0),|R|-1)}$. For GTL top-five candidate v with probability $p(v)$, the Guide selects $\arg\min_v[-\log(\max(p(v), 10^{-12})) + 0.5d_G(v, e_t)]$. Without prior route state, it uses GTL top-1. The state never crosses to the Navigator as structured data. A Guide goal confirmation is encoded as answer text; the Navigator stops by matching that received text, without a separate signal. The system executes one causal trajectory, with no replay or episode selector.

This alternative uses no shared QG, question fingerprint, target-language signature, target-description transfer, language DFS, visual signature, trajectory splicing, evaluation cache/label, manual evaluation annotation, Gemini, UniAPI, or hosted inference API.

Data, models, selection, and execution. The base is RAINbow commit `4ef165fad77e67502695 8e16d9e516e523192a33`. Frozen components are the official DST Navigator, LANA QG, LANA answer generator, and GTL localizer; no weights were trained or fine-tuned. Their upstream data provenance is RAIN/RxR/R2R/CVDN and Matterport3D. No additional dataset was used.

Top-$k = 5$ and weight 0.5 were fixed using 806 train episodes from 30 scan-disjoint scans. Relative to the same-fold base, train Score increased from 60.2713 to 64.2928 (+4.0215 points), with scan-clustered 95% CI [+2.1385,+6.0464]. Final inference used seed 0, one NVIDIA H100 GPU, Python 3.10, PyTorch 2.10, batch size 8, WTA `ctc_0.7_2`, answer-path limit 20, and a 50-action cap.

Results and limitations.

Split	Episodes	Score	SR	Steps	Dialogs
val-seen	91	0.5977	65.93%	21.59	3.26
val-unseen	241	0.2646	31.95%	24.59	5.29
released Test	285	hidden	27.37%	27.37	6.05

The released-Test result is 78/285 and is not the hidden ranking Test Score. The method was frozen from train-only evidence before its saved Test run; the official outcomes did not alter its parameters. Its main limitation is weak unseen localization: an incorrect Guide localization can still produce an incorrect route or confirmation.

Reproducibility. The delivery includes the source overlay, fixed configuration, tests, exact submission, checksums, model/data card, and runtime-weight hashes. Submission SHA256 is `52c191922ff14423efe5b290c0a9e1aa 0fa0e094c06227145db3a421fbbf5efc`.